



Artificial Intelligence (AI) Certification Program

Go beyond classical ML into Deep Learning, NLP, Computer Vision and Reinforcement Learning — and deploy real AI models on AWS & Azure.

LEARNING PATH · PROGRAM SYLLABUS · REAL-TIME PROJECTS & CASE STUDIES

350+ Hrs

Live Instructor-Led Training

Rs. 72,177

Avg. AI/ML Monthly Salary*

12

Real-Time Projects & Use Cases

300+

Hiring Partners

01 Learning Path

1

Cohort Orientation & Programming Foundations

8 Hrs

How AI & ML are reshaping industries, the role of data in decision-making, and fundamentals of programming and statistics for non-programmers.

2

Python Programming — Basic to Advanced

50 Hrs

Python fundamentals, functions & file handling through to NumPy, Pandas, Matplotlib and Seaborn — the foundation every AI model is built on.

3

Machine Learning Foundations

40 Hrs

Data preprocessing, classification & regression algorithms, ensemble methods, clustering and dimensionality reduction — the groundwork for deep learning.

4

Deep Learning — TensorFlow, Keras & PyTorch

40 Hrs

Neural networks & backpropagation, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN/LSTM), and building models across TensorFlow, Keras and PyTorch.

5

Natural Language Processing (NLP)

40 Hrs

Text pre-processing, similarity & distance algorithms, sentiment analysis, named entity recognition, and topic modelling with Latent Dirichlet Allocation.

6

Computer Vision

24 Hrs

Image processing with OpenCV, edge & feature detection, Haar cascade face/eye detection, and object localisation with Single Shot MultiBox Detectors.



7

Reinforcement Learning & AI Deployment

24 Hrs

Markov Decision Processes, Q-Learning & Policy Gradient methods with OpenAI Gym, then deploying trained AI models on AWS SageMaker and Azure Machine Learning.

02 Program Syllabus

TERM 1 · PYTHON PROGRAMMING

50 HOURS

Programming Foundations

- Environment setup, Anaconda & Jupyter Notebook
- Lists, tuples & dictionaries with built-in methods
- Strings, decisions & loop control
- Functions, modules, lambda & generators

Data Analysis & Visualization

- File I/O, exception handling & regular expressions
- Data visualisation using Matplotlib and Seaborn
- Data analysis using NumPy and Pandas
- Coding assessment plus real-world case studies

TOOLS & TECHNOLOGIES

Python

NumPy

Pandas

Matplotlib

Seaborn

TERM 2 · MACHINE LEARNING FOUNDATIONS

40 HOURS

Classification & Regression

- Data preprocessing — encoding, scaling & missing values
- K-Nearest Neighbours and Decision Tree models
- Logistic Regression and evaluation metrics
- Random Forest with hyperparameter tuning

Probabilistic & Unsupervised Models

- Naive Bayes — Gaussian, Multinomial & Bernoulli
- Principal Component Analysis (PCA)
- K-Means and Hierarchical Clustering
- Support Vector Machines (SVM)

TOOLS & TECHNOLOGIES

Scikit-Learn

GridSearchCV

TERM 3 · DEEP LEARNING (TENSORFLOW, KERAS & PYTORCH)

40 HOURS

Neural Networks & CNNs

- Neural network fundamentals, graphs & TensorBoard
- Convolutional layers, pooling & deep CNN architecture
- Backpropagation — theory and worked examples
- Building a CNN for image classification

RNNs, Keras & PyTorch

- Recurrent Neural Networks, LSTM & GRU theory
- PyTorch — tensors, autograd, linear/logistic regression
- Keras Sequential & Functional API, shared layers
- Building CNNs and RNNs in PyTorch, GPU acceleration

TOOLS & TECHNOLOGIES

TensorFlow

Keras

PyTorch

TFLearn



TERM 4 · NATURAL LANGUAGE PROCESSING

40 HOURS

Text Pre-Processing & Retrieval

- Tokenization, stop-word removal & stemming
- TF-IDF, precision/recall & F-score for retrieval
- Cosine similarity and Levenshtein distance
- KNN & K-Means for document retrieval

Sentiment, NER & Topic Modelling

- Sentiment analysis for social & web content
- POS tagging and Named Entity Recognition (NER)
- Topic modelling with Latent Dirichlet Allocation
- Use cases — toxic comment classification, summarisation

TOOLS & TECHNOLOGIES

NLTK

TF-IDF

spaCy-style NLP

TERM 5 · COMPUTER VISION

24 HOURS

Image Processing

- OpenCV installation and image drawing/rotation
- Histogram equalisation, thresholding & convolution
- Sharpening, edge detection & morphological transforms
- Gaussian/median blur and Canny edge detection

Detection & Localisation

- Haar cascade face & eye detection
- Contour properties, circle & line detection
- Watershed segmentation
- Single Shot MultiBox Detector (SSD) object localisation

TOOLS & TECHNOLOGIES

OpenCV

SSD

Haar Cascade

TERM 6 · REINFORCEMENT LEARNING & AI DEPLOYMENT

24 HOURS

Reinforcement Learning

- OpenAI Gym setup and environment basics
- Markov Decision Processes & Monte Carlo methods
- Temporal Difference learning & Policy Gradient methods
- Deep Q-Learning techniques

Deploying AI Models

- AWS SageMaker and Azure Machine Learning overview
- Data preparation, feature engineering & model training
- Creating endpoints and batch inference deployments
- Monitoring model drift and A/B testing in production

TOOLS & TECHNOLOGIES

OpenAI Gym

AWS SageMaker

Azure ML

03 Real-Time Projects & Case Studies

**SPAM Prediction
Using RNN**

Project

Classify SMS/email text as spam or not spam using a Recurrent Neural Network text model.

*Hands-on Build***Image Classifier
Using PyTorch**

Project

Build and train a CNN-based image classifier end-to-end using the PyTorch deep-learning framework.

*Hands-on Build***AI-Based Live
Face ID for
Crowds**

Capstone

A real-time face-identification system for crowd settings, built with Haar cascades and a neural classifier.

*Capstone Project***Object
Localisation with
SSD**

Project

Detect and localise objects within an image using a Single Shot MultiBox Detector in TensorFlow.

*Hands-on Build***Solving the Taxi
Environment (RL)**

Project

Train a reinforcement-learning agent to optimise pickups and drop-offs in OpenAI Gym's Taxi environment.

*Hands-on Build***Solving Frozen
Lake (RL)**

Project

Apply Q-Learning and policy methods to navigate OpenAI Gym's Frozen Lake environment safely.

*Hands-on Build***JPMorgan —
Insurance
Premium
Forecasting**

12 Hrs

Forecast tax-friendly insurance premiums and power targeted marketing using Random Forest classification.

*Project Certified***OLA — Ride
Demand
Forecasting**

18 Hrs

Build a time-period demand forecasting model using location data to cut rider waiting time.

*Project Certified***Netflix — NLP
Recommendation
Engine**

17 Hrs

Build a regional content recommendation engine from viewer activity data using NLP techniques.

*Project Certified***Saudi Aramco —
Drilling Site
Analytics**

14 Hrs

Apply big-data analytics to identify the most cost-effective, high-yield drilling sites (LWD technique).

*Project Certified***IBM — Workforce
Performance
Modelling**

19 Hrs

Identify performance variation across 14,000+ employees using regression and other ML techniques.

*Project Certified***Samsung —
Health Activity
Recognition**

13 Hrs

Continuously analyse mobile sensor data to power accurate human-activity tracking for a healthcare app.

Project Certified